

Solute recycling by crops and leaching in a drained arable soil

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Summary

Preferential flow, as it bypasses the soil matrix, can greatly enhance the leaching of chemicals. When a soil is drained there is the risk that such short-circuiting results in more or less direct passage of polluting chemicals from the soil to the groundwater. If the groundwater table is shallow the chemicals could be transferred back into the surface soil by hydraulic lift through roots and subsequent release by exudation or from decaying plant residues and again become exposed to leaching by preferential flow, thus strongly enhancing the chance of export via the drains.

We investigated the leaching of bromide in a tile-drained arable field over 2 years of crop rotation. The site was a former wetland, artificially drained a century ago for agriculture. Bromide was applied over 1.6 ha at a dosage of 10 g Br per m² in August 1995 after the harvest of wheat. During the 2 years 18% of the applied bromide was exported via the drainage system, most of it in preferential flow events and more than half of it in a single winter storm 5 months after the application. Within 7 months 56% of the applied tracer was leached out of the main root zone into the groundwater. Subsequently the tracer re-emerged in water taken up by sugar beet in the following season. The beet accumulated 50% of the initially applied bromide in their leaves and released it again after harvest when the leaves were left as green manure on the field. Our results show that this recycling of solutes to the topsoil can have an important influence on their leaching as the solutes are thus again exposed to preferential transport into drains in the course of preferential flow events.

Introduction

Various studies have demonstrated the importance of preferential flow from agricultural soils. Short-circuiting of flow through soil undermines the ability of the soil to act as a natural filter, as the matrix is largely bypassed and the residence time in the biologically active soil is reduced. This role may be particularly serious in soils with a shallow groundwater table, where drains provide artificial preferential pathways for leachates into surface waters.

Bronswijk *et al.* (1995) applied 12 g Br m⁻² to an area of 15–30 m² and monitored the concentration of bromide in the drain outflow and the groundwater. They found that although only a few per cent of the bromide passed through macropores, preferential leaching along these pathways was the cause of concentration peaks in the drain discharge. In accord with this they also found that in summer, when the soil was dry and

cracked, more bromide was leached than in winter, when the clayey soil was wet and swollen. Other authors have obtained similar results for other soils where macropore flow contributed to the fast arrival of bromide, nitrate or chloride in drain outflow (Magesan *et al.*, 1995; Bjorneberg *et al.*, 1996; Villholth *et al.*, 1998; Larsson *et al.*, 1999).

When preferential flow and transport processes are assessed by tracer experiments under field conditions, plant root activity must be taken into account as a major factor. In addition to their effect on the soil water regime, plants may also have an important direct impact on the exposure of solutes to preferential flow. Schnabel *et al.* (1995) found in a tracer experiment with bromide that ryegrass accumulated an average of 32% of the applied bromide in the aboveground biomass. Owens *et al.* (1985), Fransi *et al.* (1987), Kung (1990) and Steenhuis *et al.* (1990) also reported large uptakes of bromide by plants. If a solute taken up by roots is removed from the system in harvested product, then this uptake simply acts as an irreversible sink similar to radioactive decay or biodegradation. If the

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solute is released again into the soil, for example by decomposition of roots or crop residues left on the soil surface as well as exudation from roots, then the solute will redistribute in the soil profile. In this way, the solute, which had already been leached into the subsoil, can be recycled back to the topsoil from where it may be leached again. This recycling can be particularly important in soils where plant roots are hydraulically connected to a shallow groundwater table.

Solutes released from decomposing plants and their roots, as well as ones exuded from roots, are as prone to leaching by preferential flow as those applied to the soil surface. Stewart *et al.* (1999) studied the spatial relation between roots and macropores and found that the distribution of roots in the vicinity of macropores is larger than what could be expected by a mere random distribution. Bundt *et al.* (2001), studying microscale dynamics of carbon and nitrogen in a forest soil, found that the soil organic matter was younger and N cycling more rapid in preferential flow paths than in the soil matrix. These findings suggest a close association between preferential flow paths and plant roots. Enrichment of solutes in the soil matrix along roots would subsequently lead to increased concentrations in preferential flow channels. Sinaj *et al.* (2002) showed that phosphorus leaching through grassland soils was linked to increased concentrations in pathways of preferential flow.

Combining the available evidence we hypothesize that in soils with a shallow groundwater table solute recycling from the subsoil into the topsoil via plants influences leaching into surface waters via subsurface drains. Many investigators have studied solute leaching and preferential flow in field soils and many others have dealt with solute extraction from soils by plants. However, we know of no studies in which solute recycling from subsoil and shallow groundwater by crop plants has been investigated with respect to the potential role of this process in the leaching of solutes into subsurface drains, in particular by preferential flow. In this paper we describe a field tracer experiment in which we monitored the displacement of bromide in an arable soil with a shallow groundwater table for 2 years.

Materials and methods

The study site was a former wetland in a flat valley plain reclaimed for intensive agriculture at the beginning of the 20th century near the town of Otelfingen in the canton of Zürich, Switzerland (8°24.6'E longitude, 47°27.3'N latitude). We chose this site because we suspected it to be a major source of nitrate pollution in the creek receiving the drainage water. In addition, preliminary dye tracer experiments indicated that preferential flow through the drained sites played an important role in the displacement of solutes in this soil. Within the site we chose a 1.6-ha field representing a compartment drained by a single collector for the tracer experiment.

The field drains were clay tubes 1 m below the soil surface, and 20 m apart. The collector was at a depth of 1.5 m and consisted of perforated polyethylene tube with a gravel backfill (Figure 1). The soil is a Mollic Gleysol, rich in carbonate, organic matter and silt (Table 1). The land was used in an 8-year crop rotation scheme. Table 2 provides pertinent information on the land management during the experiment.

Hydrological measurements

The drainage outflow leaving the experimental field was collected in a manhole where the discharge was pumped through a flow meter. The meter and the submerged pump were calibrated such that the error in the measured cumulative discharge should not exceed $\pm 2\%$ of the true value.

The groundwater table was measured with a piezometer equipped with a pressure transducer connected to a data logger. The piezometer was placed midway between two field drains to measure the apex of the groundwater table in the drained field. Prior to our installing the automatic piezometer, we had 31 piezometers recorded manually and compared their relationship to choose the location of the reference piezometer.

Air humidity, temperature, radiation, precipitation and wind speed were recorded at a mobile meteorological station on site close to the experiment. We determined evapotranspiration by multiplying a reference evapotranspiration with crop-specific coefficients taken from DVWK (1996) (Figure 2). The reference evapotranspiration ET_0 [mm] was determined from the combination equation of Allen *et al.* (1994):

$$ET_0 = \frac{s}{s + \gamma(1 + 0.34v_2)} \left\{ \frac{R_n}{L} + \left(\frac{e_s}{s} \right) \frac{3.75\gamma}{T + 273} t v_2 \left(1 - \frac{U}{100} \right) \right\}, \quad (1)$$

where R_n [$J\ cm^{-2}$] is the net radiation at the crop surface, and t [hours] is the corresponding time. In our case ET_0 was

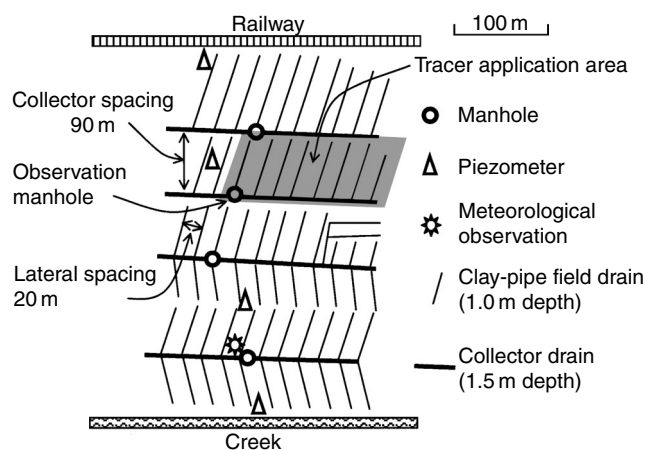


Figure 1 Schematic of the experimental site.

Table 1 Soil profile data of the experimental field

Depth /m	Bulk density ^a /g cm ⁻³	CaCO ₃	Organic matter	Clay	Silt	Sand
		/%				
0–0.25	0.70 ± 0.01	22	9	37	45	18
0.25–0.40	0.80 ± 0.09	78	10	17	73	10
0.40–1.00	1.24 ± 0.01	55	0.6	22	67	11
> 1.00	1.23 ± 0.01	45	0.9	21	62	17

^aMean and standard error.

calculated on a daily basis, thus $t = 24$ [hours], T [°C] is the average temperature at 2 m height, U is the relative vapour pressure deficit, v_2 [m s⁻¹] is the wind speed at 2 m height, s [kPa K⁻¹] is the slope of the vapour pressure curve, γ [kPa K⁻¹] is the psychrometric constant, L [J cm⁻² mm⁻¹] is the relative evaporation heat of water, and e_s [kPa] is the saturated vapour pressure.

We tested this method using a 4-year time series of meteorological and lysimeter data collected at the agricultural research station in Zürich situated only 4 km east of the study site in the same valley. The agreement between the calculated cumulative evapotranspiration and the experimental data obtained from the water balance of the lysimeter was very good, as illustrated in Figure 2. Only in autumn 1984 did the calculated evapotranspiration underestimate the measurements of the lysimeter. During the remaining 4 years with different crops the cumulative evapotranspiration followed the 1:1 line in Figure 2. Thus, we considered the method to be valid to predict the evapotranspiration on the study site.

Bromide application

A dose of 10 g Br m⁻² was applied as a 2.36-M KBr solution under wind-free conditions to the experimental area on 23 August 1995, with a spray-trailer pulled by a tractor. The application accuracy of the spray-trailer according to the manufacturer should be within ±5% of the true value. Replicate samples of the applied tracer measured by absorbent sheets gave a mean of 10 ± 1.4 g bromide m⁻².

Table 2 History of the agricultural management of the experimental field during the tracer experiment

Date	Status of the crop
9 August 1995	Harvest of winter wheat
11 August 1995	Sunflowers seeded
23 August 1995 ^a	Sunflowers emerged and about 5 cm tall
6 November 1995	Sunflowers killed by frost
30 March 1996	Residues of the sunflowers ploughed into soil and sugar beet seeded
25 September 1996	Sugar beet harvested
8 November 1996	Seeding of winter wheat after ploughing and soil preparation
11 August 1997	Harvest of winter wheat

^aTracer application.

Sampling and analysis of drainage water

We collected samples of the drainage water in 500 ml bottles using a 24-position ISCO sampler (model 2900) connected to a data logger. Sampling started when the discharge reached a critical flow rate of 0.22 l s⁻¹. From each bottle, a subsample of 10 ml was transferred into a small PVC bottle and frozen for storage until analysis. After thawing, the samples were filtered through 0.25 µm cellulose-acetate filters and then analysed by Dionex anion chromatography.

Plant sampling and analysis

Sunflowers were sampled on 10 October 1995, 7 weeks after application of the tracer. Plants were collected from 13 sub-areas of 0.4 × 0.4 m randomly selected from all over the field. We collected all the aboveground parts within each subarea by cutting the shoots directly above the ground. Sugar beet plants were sampled on 4 September 1996. Three randomly chosen plants were dug out of the ground and divided into leaves and roots. We sampled the wheat on 30 June 1997 from three 1.0 × 1.0 m subareas by cutting the plants directly above the ground and separating ears and stems.

In the laboratory the plant samples were chopped, oven-dried for 24 hours at 60°C and weighed. Ground subsamples of 1 g each were shaken for 30 minutes in 50 ml of a 0.5% trichloroacetic acid solution. After 1:10 dilution with distilled water, the extracts were analysed for bromide by Dionex anion chromatography.

Soil sampling and analysis

Soil cores were sampled at random on 4 October 1995 (five cores), 15 May 1996 (four cores), 25 February 1997 (three cores) and 30 June 1997 (three cores). The samples, which had a diameter of 5 cm, were taken by means of a hollow auger to a depth of 125 cm in 25-cm-long sections. The first three 25-cm sections of each core were further subdivided into 12.5-cm sections in the laboratory.

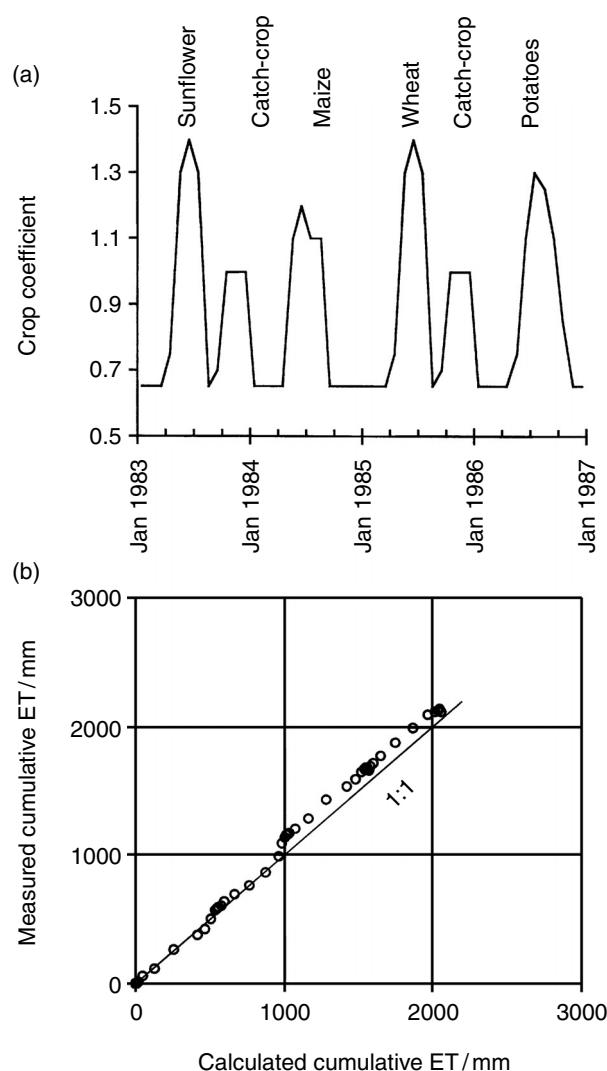


Figure 2 (a) Crop coefficients used to modify the reference evapotranspiration. (b) Comparison of evapotranspiration (ET) measured with a lysimeter and calculated with the equation of Allen *et al.* (1994).

The samples were oven-dried for at least 24 hours at 105°C. Fifty-gram aliquots of the dry samples were suspended in 100 ml distilled water each and shaken for 2 hours. The extracts were filtered through 0.25 µm cellulose-acetate filters and, after the first 10 ml of each filtrate had been discarded, were analysed by anion chromatography in the same way as the drain water samples.

We calculated the bromide concentration of the soil water by dividing the mass of bromide analysed in the bulk soil samples by the water content of the samples given by the weight decrease before and after drying. Bromide contents per unit of bulk soil volume were calculated from the soil bulk density values of the reference profile given in Table 1. Each of these values represents the average of three replicate samples with a volume of 1 litre.

Results

Water regime and bromide export by drains

From 23 August 1995 until 30 July 1997 precipitation amounted to 1682 mm, while the calculated actual evapotranspiration was 1152 mm, giving a net infiltration of 530 mm. Since the experimental field was in a flat valley there was no runoff. When the rainfall exceeded the infiltration capacity of the soil the surplus water lay ponded on the soil surface. Over the same period, drainage outflow accumulated to 748 mm. Figure 3 shows that up to January 1997 drain outflow agreed well with the net infiltration and that the two diverged only during the last few months of the observation period. In spring 1997 precipitation did not meet the demand of the growing winter wheat, and evapotranspiration resulted in a net depletion of the soil water and subsequently a fall in the groundwater table. By 30 June 1997 the calculated water deficit was 218 mm. Figure 3 shows that the groundwater table dropped much deeper in the spring of 1997 than in 1996. In May and June of 1997, which were extraordinarily dry, the groundwater table reached levels more than 0.5 m below the collector drain.

Figure 4 shows the cumulative bromide export by drainage. From the time of application in August 1995 until July 1997 we collected a total of 17.9% of the applied bromide at the drains. About half of this export occurred during a single event in December 1995, when within 44 hours, 53 mm of rain fell at a maximum rate of 3.2 mm hour⁻¹ on an almost saturated soil. The groundwater table was 0.6 m below soil surface the day before the event started, and soil water tension was 1.9 kPa in the topsoil 3 days earlier when the last measurement was taken before the event. Abbaspour *et al.* (2001) analysed the time courses of bromide concentration and flow rate of the drainage discharge in a modelling study and concluded that the concentration peaks during infiltration were due to leaching by preferential flow (Figure 4).

Bromide displacement in the soil profile and into the groundwater

Spatial variation also led to a large uncertainty in the estimates of the total mass of bromide stored in the soil profile (Table 3). The analytical variation of the measured concentration of the soil samples was determined by duplicate analysis of the same soil samples. A set of 11 samples was analysed twice and resulted in a coefficient of variation of 12% for the measured content of bromide in the soil samples. The variation due to the sampling and analysis, although considerable, was still much smaller than the variation between bromide contents of different soil cores at the four sampling dates given in Table 3. This additional variation can be attributed to the variation of the application as well as to the natural heterogeneity of all the factors influencing the vertical distribution of surface-applied bromide in the soil profile.

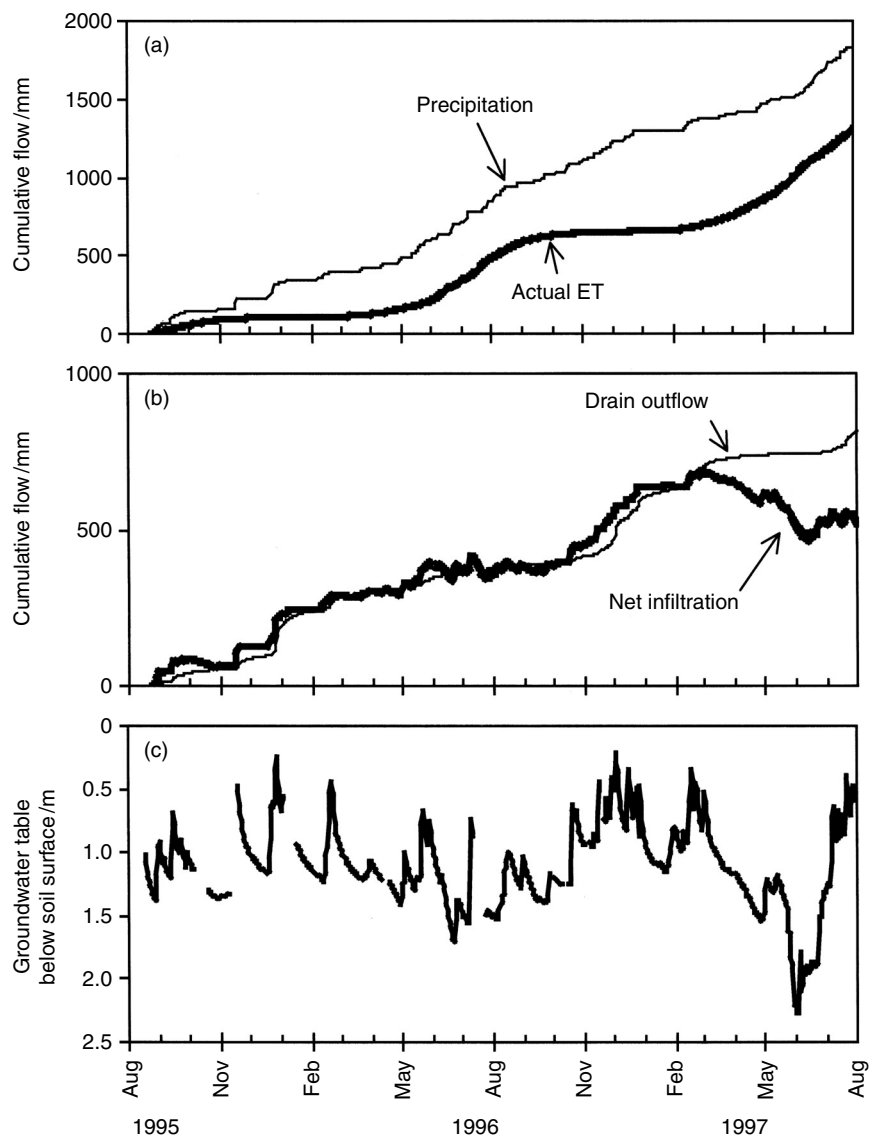


Figure 3 (a) Cumulative precipitation and evapotranspiration; (b) net infiltration and drain discharge after tracer application on 23 August 1995 until 30 July 1997; and (c) the observed depth of groundwater table below the soil surface for the same period of observation.

Table 4 lists the cumulative net infiltration after application of tracer on 23 August 1995 until the sampling dates. The expected penetration depth represents the depth where the front of cumulative net infiltration would have reached if the infiltration had occurred uniformly at constant water content distribution in the manner of piston flow. Note that in this notion only the region above this front is assumed to meet the evapotranspiration requirements. The error in this assumption was negligible because the infiltration of the tracer solution started almost at the same time as the root growth of the emerging sunflower seedlings, and also because the displacement of the tracer through the upper part of the profile occurred mainly during the non-growing season. As a reference state we chose volumetric water contents of 0.42 in the topsoil (0–0.25 m), 0.51 in the silty layer (0.25–0.40 m) and 0.50 in the loamy subsoil below. This distribution of water content was observed in the field on 12 December 1995 when the soil

had drained for several days and there was negligible evapotranspiration.

Table 3 lists the depth distributions of the bromide, calculated from the bulk density of the sampled soil layers, while in Figure 5 the respective profiles of bromide concentrations in the soil water phase are illustrated for the soil sampling dates. These concentration profiles show that 6 weeks after the application, on 4 October 1995, there was a single, broad peak rising from the surface to a maximum at 30 cm depth, then sharply declining between 50 and 60 cm and then extending into a long advancing tail below 1 m depth. Cumulative net infiltration from the time of application of the tracer was 77 mm, corresponding to a penetration depth of 0.18 m (Table 4). Thus the actual vertical displacement of the bromide was on average more than twice as deep as expected if all the pore water had participated in the transport.

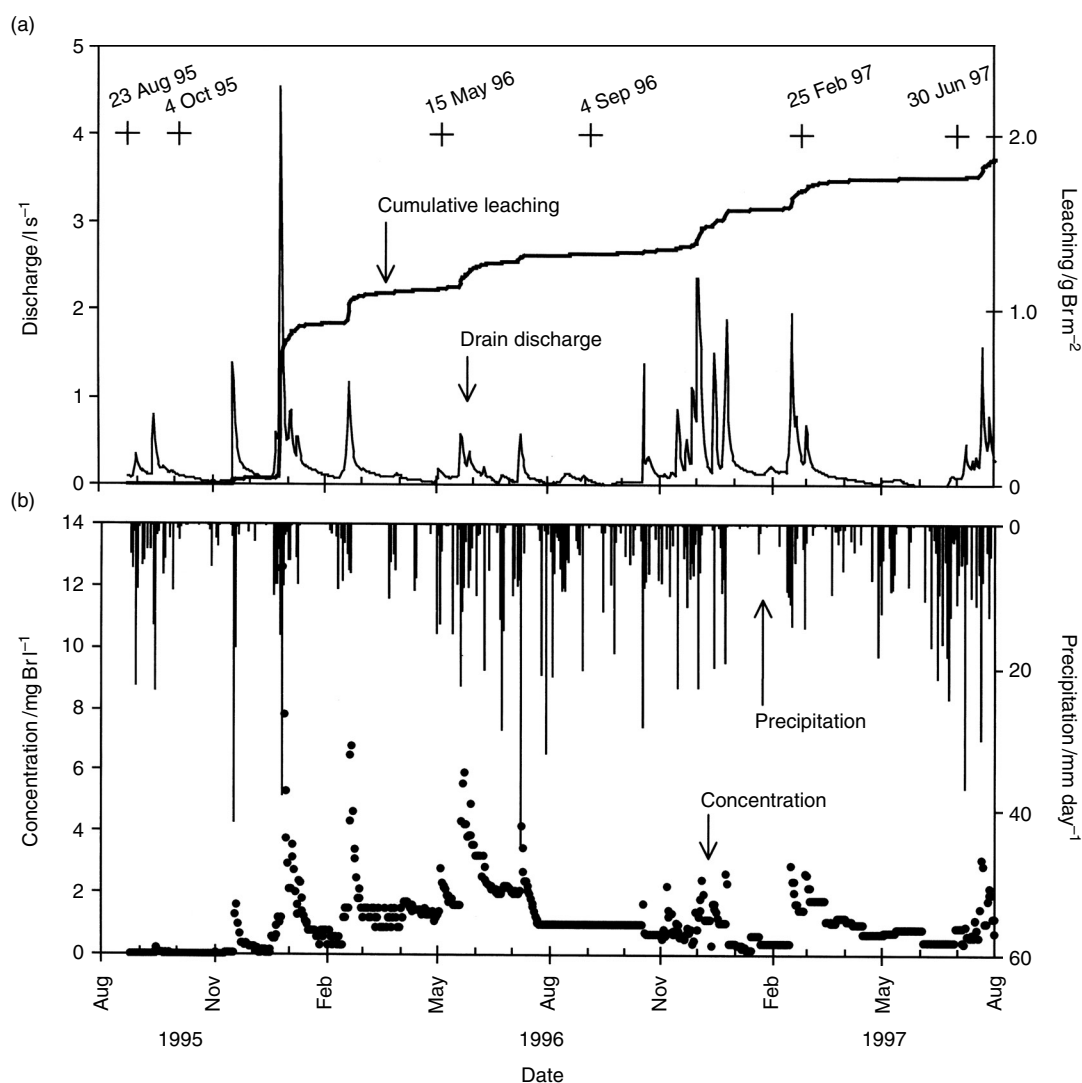


Figure 4 (a) Cumulative bromide export via drainage and drainage discharge. (b) Bromide concentration of the drainage water and daily precipitation. +, Date of sampling.

Table 3 Average and standard error of bromide (g Br m^{-2}) in different soil layers relative to bulk soil volume, measured at four dates. 10 g Br m^{-2} was applied on 23 August 1995

Layer /cm	4 October 1995	15 May 1996	25 February 1997	30 June 1997
0–12.5	0.51 ± 0.16	0.17 ± 0.05	0.60 ± 0.30	0.53 ± 0.10
12.5–25.0	0.74 ± 0.18	0.24 ± 0.03	0.50 ± 0.18	0.45 ± 0.06
25.0–37.5	1.27 ± 0.38	0.40 ± 0.17	0.88 ± 0.38	0.41 ± 0.09
37.5–50.0	1.64 ± 0.71	0.56 ± 0.35	1.16 ± 0.47	0.32 ± 0.11
50.0–62.5	0.35 ± 0.22	0.46 ± 0.19	0.81 ± 0.39	0.15 ± 0.02
62.5–75.0	0.10 ± 0.03	0.58 ± 0.16	0.70 ± 0.24	0.14 ± 0.03
75.0–100	0.10 ± 0.02	0.61 ± 0.12	1.54 ± 0.59	0.19 ± 0.042
100–125	0.05 ± 0.01	0.29 ± 0.06	0.48 ± 0.14	0.00 ± 0.00
Total mass	4.76 ± 1.44	3.32 ± 0.96	6.67 ± 2.03	2.19 ± 0.39
Number of samples	5	4	3	3

Table 4 Cumulative net infiltration after bromide application on 23 August 1995 and the expected depth of the bromide front assuming piston flow

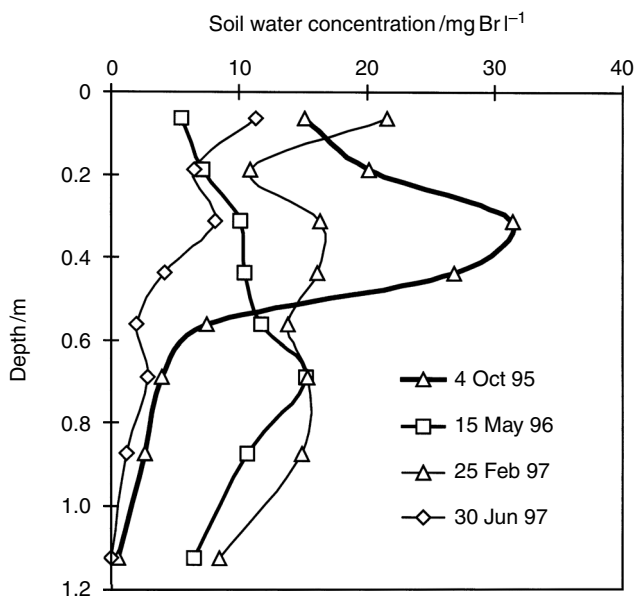
Date	Net infiltration /mm	Expected penetration depth /mm
4 October 1995	77	183
15 May 1996	337	710
4 September 1996	377	792
25 February 1997	673	1383
30 June 1997	530	1096

By 15 May 1996, 337 mm of water had infiltrated. The bromide distribution had flattened, extending over the entire soil profile with a maximum occurring around 0.70 m depth. Note that the depth of the concentration maximum coincides with the calculated depth of penetration by equivalent piston flow of 0.71 m in Table 4. This might be a mere coincidence, however, and does not mean that the vertical movement of bromide had slowed down. By 15 May 1996 we had recovered in the soil only one third of the mass of the bromide applied, i.e. 3.32 g Br m^{-2} , whereas 1.12 g Br m^{-2} had already been exported through the drainage system (Table 6). Thus, more than half of the applied mass of bromide (5.56 g Br m^{-2}) was apparently in the saturated zone below the sampling depth of 1.25 m (cf. Figure 3).

Some retardation in the vertical tracer movement could have resulted from the uptake by the sunflowers and a subsequent release through decomposition over the winter. Absence of a second concentration peak at the soil surface in the sampling

of 15 May 1996 might have been due to mixing by hydrodynamic dispersion and ploughing during the time between release and sampling. A strong concentration increase was seen in the uppermost layer at the time of the third soil sampling on 25 February 1997 despite ploughing and considerable infiltration over the winter after harvest (Figure 5). By the last sampling on 30 June 1997, bromide concentrations had decreased again in the entire profile, but much less in the topsoil than in the subsoil, leading to a clear concentration maximum at the soil surface (Figure 5). In the lower part of the profile, the decrease corresponded to a depletion of bromide mass. In the topsoil (0–0.25 m) the mass was approximately the same as in the previous sampling (Table 3). This means that no net leaching from the topsoil into the subsoil had occurred. Indeed, a water balance calculation showed that the water lost by evapotranspiration considerably exceeded precipitation during the period between the last two samplings, leading to a corresponding drop in the groundwater table (Figure 3). Thus, until the end of May 1997 when rainfall again started to reverse this trend, water movement in the profile was dominated by capillary rise. The small concentration in the lower part of the subsoil suggests that the bromide that was carried upwards with the flow was not replaced by bromide from the groundwater.

At the beginning of the experiment, sunflowers were grown as a catch-crop after wheat. The seedlings had just emerged when the tracer was applied on 23 August 1995. At the first sampling (10 October 1995), $1.32 \pm 0.17 \text{ g}$ bromide per m^2 had been taken up into the leaves and shoots (Table 5). This fraction was returned into the soil when the plants died from frost in November 1995 and were ploughed into the soil just before sugar beets were sown in March 1996. On 4 September 1996, shortly before the sugar beet was harvested, $5.08 \pm 2.10 \text{ g Br m}^{-2}$ of the applied mass of bromide was found in the beet, mostly in the leaves ($4.97 \pm 2.00 \text{ g Br m}^{-2}$) and only a small amount ($0.11 \pm 0.07 \text{ g Br m}^{-2}$) in the roots. As with the sunflower, the sugar beet also recycled bromide to the soil surface. In this case the amount of bromide released into the soil by decomposition was much larger, representing around half of the bromide originally applied. The wheat grown in 1997 accumulated $0.68 \pm 0.14 \text{ g Br m}^{-2}$ in the stalks and leaves, and $0.07 \pm 0.01 \text{ g Br m}^{-2}$ in the grain. The large uncertainty

**Figure 5** Concentration of bromide in the soil water solution versus depth at four sampling dates.**Table 5** Averages and standard errors of bromide uptake by the crops (g Br m^{-2})

Crop	Sunflower	Sugar beet	Wheat
Date of sampling	10 October 1995	4 September 1996	30 June 1997
Roots	—	0.11 ± 0.07	—
Grain	—	—	0.07 ± 0.01
Leaves	1.32 ± 0.17	4.97 ± 2.00	0.68 ± 0.14
Number of samples	13	3	3

of these figures results primarily from the large variation of bromide contents in the plant samples.

Discussion and conclusion

Table 6 shows that large fractions of the applied bromide must have been transferred back and forth through soil, plants and the groundwater system. The bromide uptake by the sugar beet was found to be about 50% of the initially applied tracer mass. Because the uptake by the sugar beets exceeded the mass of bromide stored in the soil profile at the beginning of the growing season (3.32 g Br m^{-2} on 15 May 1996) we conclude that a fraction of the accumulated bromide originated from the groundwater below.

When the leaves of the beet were worked into the soil after harvest, all the bromide taken up, except for the tiny fraction exported with the harvest of the roots, was returned into the plough layer. The precipitation over the following winter was sufficiently large to distribute the released bromide deep down into the subsoil, as observed in the sampling of February 1997 (Figure 5).

In the following months the soil profile again lost almost all of the mass of bromide that it had gained from the incorporation of the residues. Since the water balance from 25 February 1997 to 30 June 1997 was negative, with a deficit of more than 140 mm (Table 4), leaching and deep seepage through the soil matrix into the groundwater could not have been plausible. Also, the uptake of bromide by the growing wheat was not sufficiently large to account for this loss.

A more likely explanation is leaching with bypass flow. We may notice that in the last weeks before the sampling heavy rain had occurred after a long period of drought, leading to a marked rise in the groundwater table (see Figures 3 and 4). As this precipitation infiltrated into an initially dry soil it can be expected that a large part of this water bypassed the soil matrix below the plough layer as preferential flow, leaching bromide more or less directly into the groundwater zone. As long as the groundwater table was still below the drains, drainage discharge did not capture this bromide.

Large accumulations of bromide by crop plants have also been reported by other investigators. Owens *et al.* (1985) found an uptake of 5.4 g Br m^{-2} by grass, corresponding to 32% of the applied mass, Kung (1990) found 6 g Br m^{-2} in potatoes representing 53% of their application, Steenhuis *et al.* (1990) found 3.6 g Br m^{-2} in maize corresponding to 30% of the application, and Fransi *et al.* (1987) found 8.6 g Br m^{-2} in wheat representing 8.6% of the application. The last figure is in good agreement with the uptake by wheat in our case of 11% of the bromide initially present in the entire soil profile, while the uptake of 50% of the bromide in the subsurface by the sugar beet is comparable to the uptake observed in potatoes by Kung (1990). In contrast, Schulin (1993) found no bromide extraction by plants when the tracer had been leached into a gravelly subsoil below the root zone prior to the start of plant growth. It appears that the timing of tracer availability in the root zone, plant growth stage and rooting depth, and thus the transpiration activity of the plant influence the amount of solute taken up by plants.

Compared with the rates of bromide transfer between soil, plants and groundwater, export with drainage discharge was of secondary importance. Half of this loss, amounting to 18% of the applied mass over 23 months from August 1995 until June 1997, occurred through a single event in December 1995. If seepage had been uniform then the bulk of the applied tracer should not have reached the drains. This observation, along with the fact that bromide concentration in the drainage sharply increased with peak flow and then dropped again to the previous concentration in the base flow, indicated that preferential flow was the dominant mechanism of transport. Indeed, as mentioned before, Abbaspour *et al.* (2001) showed that the observed pattern of bromide discharge could be explained by a two-domain model separating slow matrix and fast macroporous flow, but not by a single-domain model of Darcian flow and convective-dispersive transport. Other authors also interpreted attributed synchronous peaks in tracer concentrations and tile discharge rates (Figure 4) as an indication of transport by preferential flow (Magesan *et al.*, 1995; Lennartz *et al.*, 1999; Kung *et al.*, 2000).

Table 6 Mass distribution (g Br m^{-2}) of the applied bromide tracer over the five sampling times. 10 g Br m^{-2} was applied on 23 August 1995

Compartment	Sampling date				
	10 October 1995	15 May 1996	4 September 1996	25 February 1997	30 June 1997
Crop remaining on the field	1.32 ± 0.17	—	4.97 ± 2.00	—	0.68 ± 0.14
Cumulative export with crop	—	—	0.11 ± 0.07	0.11 ± 0.07	0.18 ± 0.08
Soil: 0–1.25 m	4.76 ± 1.44	3.32 ± 0.96	^b	6.67 ± 2.03	2.19 ± 0.39
Export via drain	0.00	1.12 ± 0.13	1.32 ± 0.15	1.68 ± 0.19	1.76 ± 0.21
Groundwater ^a	3.92 ± 1.50	5.56 ± 0.97	^b	1.54 ± 2.04	5.19 ± 0.48

^aEstimated as the difference between tracer application and recovery in the compartments.

^bNo soil sampling was possible.

The fact that most of the tracer export by the drainage system was associated with a few heavy rain events, while only very little tracer was exported with base flow, precludes generalizing conclusions with respect to the quantitative role of these fluxes over a long time period. However, the results clearly show that such events can play a dominant role, not only for leaching during the growing season, but also at other times. Our results also show that crops can recycle substantial amounts of bromide that were already leached into the groundwater back to the topsoil where they were again exposed to preferential flow. Thus, even 2 years after application the tracer was still subject to the risk of being leached into the drains by preferential flow.

While bromide is assumed to mimic the physical transport behaviour of nitrate, root uptake of the two solutes may be quite different. Nonetheless we can conclude that nitrogen could also be extracted in the form of nitrate from shallow groundwater under similar conditions and returned to the topsoil through mineralization of plant residues. While plant uptake reduces groundwater contamination, mineralization of residues may again increase the chance of preferential leaching. With nitrogen, more complicated processes can occur because other pathways such as immobilization by microbes and denitrification will reduce the chance of leaching by reducing the available nitrate. Although tracer experiments with bromide cannot resolve the role of these biological processes, they may be used to assess the role of the physical processes and to estimate transfer times in the various subsurface compartments, which in turn determine the exposure to biological transformation processes.

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